

Mohammed Ahmed Moawed

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Professional Summary

Motivated Information Technology student at Cairo University with a strong foundation in software engineering and full-stack development, currently specializing in Cloud computing and DevOps engineering. Actively expanding expertise in cloud infrastructure, containerization, and automation through the Digital Egypt Pioneers Initiative (DEPI). Proficient in leveraging tools like AWS, Docker, Kubernetes, and Ansible to build scalable, reliable systems. Passionate about automating CI/CD pipelines, system monitoring with Prometheus, and bridging the gap between development and IT operations.

Education

Cairo University | Giza, Egypt

Bachelor of Computers and Artificial Intelligence (Information Technology)

2023 – Present

- Third-year student majoring in Information Technology.

Technical Skills

- **Cloud & DevOps:** AWS, Docker, Kubernetes, Ansible, Prometheus, CI/CD Pipelines, Containers.
- **Operating Systems & Scripting:** Linux Administration, Bash/Shell Scripting, SSH.
- **Version Control & Collaboration:** Git, GitHub, Git workflows.
- **Programming Languages:** C++, JavaScript, TypeScript, Python (Basic), Java (Academic), C# (Academic).
- **Web Development:** MERN Stack (MongoDB, Express.js, React, Node.js), HTML, CSS, TailwindCSS.
- **Databases:** MongoDB, SQL.
- **Software Engineering:** OOP, SOLID Principles, UML Diagrams, System Design Patterns.

Training & Programs

Digital Egypt Pioneers Initiative (DEPI) – DevOps Track

Intensive 180-hour hands-on training program focused on modern Cloud and DevOps

practices.

- **Cloud Infrastructure:** Gained practical experience deploying and managing resources on AWS.
- **Containerization & Orchestration:** Built and managed isolated environments using Docker and orchestrated container deployments at scale using Kubernetes.
- **Automation & CI/CD:** Automated server provisioning and configuration management using Ansible, and implemented continuous integration and delivery concepts.
- **Monitoring & Linux:** Managed Linux-based servers using Bash scripting and explored system observability and metrics monitoring using Prometheus.

Software Developer Intern for Elevvo

- Collaborated on developing scalable web solutions, focusing on writing clean, maintainable code and participating in the full software development life cycle.
- Applied modern development practices and version control to deliver features, ensuring alignment with project requirements and performance standards.
- **Project Link:** github.com/Mohammed-Bahr/Elevvo-Internship

Projects

Food Platform (MERN Stack) | [GitHub Link](#)

- Designed and implemented a robust backend using Node.js, Express, and MongoDB.
- Created structured data models for Users, Recipes, Favorites, and Categories.
- Applied RESTful API principles, JWT-based security, and basic authentication workflows.
- *(Optional: If you plan to containerize this, add: "Currently working on containerizing the application using Docker and setting up a CI/CD pipeline for automated testing and deployment.")*

Database Management & Data Analysis | [GitHub Repository](#)

- **SQL & Relational Databases:** Developed complex queries, managed schemas, and applied data normalization techniques using MySQL.
- **Data Engineering Foundations:** Performed extensive data cleaning and preprocessing using **Pandas** and **NumPy** to prepare raw datasets for analysis.
- **Data Visualization:** Created insightful plots and dashboards using **Matplotlib** to communicate technical findings effectively.
- **Automation:** Scripted data cleaning tasks to improve workflow efficiency, bridging the gap between raw data and actionable insights.

CPU Scheduler Simulator : ([GitHub link](#))

Developed a comprehensive Java-based simulation of core OS scheduling algorithms, including Preemptive SJF, Round Robin, Priority, and a custom **AG (Adaptive General) Scheduling** algorithm. The system models complex process behaviors such as context switching, dynamic quantum adjustment, and starvation avoidance, providing detailed performance metrics like Turnaround and Waiting times to analyze system efficiency.

Java Terminal Emulator : ([GitHub link](#))

Engineered a functional Unix-style shell environment in Java using the NIO framework to perform advanced file system operations and directory navigation. The emulator features a custom command parser supporting argument quoting, recursive file operations (copying and zipping), and output redirection (> and >>), demonstrating a deep understanding of system-level I/O and process management.

Discrete Event Car Wash Simulation :([GitHub link](#))

Designed and implemented a modular discrete-event simulation (DES) in Java to model queuing theory and system throughput in a multi-bay service facility. The project utilizes object-oriented principles to track real-time metrics such as bay utilization and average wait times, providing a customizable framework for studying the impact of arrival rates and service capacities on system performance.

Vole Machine Simulator (C++): Built a low-level simulator modeling computer memory, registers, ALU, PC, and IR. ([GitHub Link](#))

- Implemented the core instruction fetch-decode-execute cycle.
- Designed a visual display of the system state, including registers and memory mapping after execution.

Data Structures & Algorithms Repository | [GitHub Link](#)

- Implemented complex data structures and algorithms using C++ for high performance and efficiency. Includes compression methods for images and text using Vector Quantization.

C++ Game Development ([GitHub Link](#))

- Developed classic games including Snake and multiple versions of Tic-Tac-Toe with varying features and logic mechanics.

Soft Skills

- Strong problem-solving and analytical thinking
- Adaptability and rapid self-learning
- Cross-functional teamwork and collaboration

- Time management and task prioritization

Languages

- **Arabic:** Native - **English:** Very Good - **German:** Basics